

Hongchao Zhang

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I am a Postdoctoral Scholar at Vanderbilt University with a Ph.D. in Electrical Engineering from Washington University in St. Louis (2025). My research focuses on **trustworthy AI-enabled robotic systems**, integrating control theory, neural network verification, and security to provide end-to-end certification from algorithm to deployment. I develop (1) exact neural safety certificates, (2) fault-tolerant control under sensor faults and adversarial attacks, and (3) control-theoretic approaches to CPS software security. I have published 15 peer-reviewed papers at ACM CCS, NeurIPS (2), IEEE TAC, ICRA (2), and CDC (6), and received the GM AutoDriving Security Award.

EDUCATION

Washington University in St. Louis <i>Ph.D., Electrical Engineering (Advisor: Prof. Andrew Clark)</i> Thesis: "Resilient Safe Learning of Autonomous Systems"	St. Louis, MO 2020 – 2025
Worcester Polytechnic Institute <i>M.S., Electrical & Computer Engineering</i>	Worcester, MA 2018 – 2020
Nanjing University of Aeronautics and Astronautics <i>B.E., Automation Engineering</i>	Nanjing, China 2014 – 2018

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar <i>Vanderbilt University, Institute for Software Integrated Systems</i> Research on intersection among AI, CPS, and formal methods with Prof. Taylor T. Johnson, Prof. Meiyi Ma and Prof. Ahmed F. Taha.	Sep 2025 – Present Nashville, TN
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PUBLICATIONS

Journal Papers

- [J1] **[IEEE TAC] Zhang, Hongchao**, Zhouchi Li, and Andrew Clark. "Safe Control for Nonlinear Systems under Faults and Attacks via Control Barrier Functions." *IEEE Transactions on Automatic Control*, vol. 70, no. 11, pp. 7334-7349, Nov. 2025, doi: 10.1109/TAC.2025.3571338.
- [J2] **[IEEE TAC] Zhang, Hongchao**, Manan Tayal, James Cox, Pushpak Jagtap, Shishir Kolathaya, and Andrew Clark. "Stochastic Neural Control Barrier Functions." *IEEE Transactions on Automatic Control*, under review.
- [J3] **[IEEE TAC] Zhang, Hongchao**, Mohamad H. Kazma, Meiyi Ma, Taylor T. Johnson, and Ahmad F. Taha. "Verification and Forward Invariance of Control Barrier Functions for Differential-Algebraic Systems." *IEEE Transactions on Automatic Control*, under review.

Conference Papers

- [C1] **[ACM CCS] Wang, Jinwen***, **Hongchao Zhang***, Chuanrui Jiang, Andrew Clark, and Ning Zhang. "ConTest: Taming the Cyber-physical Input Space in Fuzz Testing with Control Theory." In *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2025.
- [C2] **[NeurIPS] Zhang, Hongchao**, Zhizhen Qin, Sicun Gao, and Andrew Clark. "SEEV: Synthesis with Efficient Exact Verification for ReLU Neural Barrier Functions." In *Advances in Neural Information Processing Systems*, vol. 37, 2024.
- [C3] **[IEEE CDC] Dai, Hongkai**, Chuanrui Jiang, **Hongchao Zhang**, and Andrew Clark. "Verification and Synthesis of Compatible Control Lyapunov and Control Barrier Functions." In *IEEE Conference on Decision and Control*, 8178–8185, 2024.
- [C4] **[IEEE CDC] Tayal, Manan**, **Hongchao Zhang**, Pushpak Jagtap, Andrew Clark, and Shishir Kolathaya. "Learning a Formally Verified Control Barrier Function in Stochastic Environment." In *IEEE Conference on Decision and Control*, 4098–4104, 2024.
- [C5] **[IEEE ICAA] Niu, Luyao**, **Hongchao Zhang**, Dinuka Sahabandu, Bhaskar Ramasubramanian, Andrew Clark, and Radha Poovendran. "Who is Responsible? Explaining Safety Violations in Multi-Agent Cyber-Physical Systems." In *IEEE International Conference on Assured Autonomy*, 2024.

- [C6] **[IEEE ICRA] Zhang, Hongchao**, Luyao Niu, Andrew Clark, and Radha Poovendran. "Fault Tolerant Neural Control Barrier Functions for Robotic Systems under Sensor Faults and Attacks." In *IEEE International Conference on Robotics and Automation*, 9901–9907, 2024.
- [C7] **[NeurIPS] Zhang, Hongchao**, Junlin Wu, Yevgeniy Vorobeychik, and Andrew Clark. "Exact Verification of ReLU Neural Control Barrier Functions." In *Advances in Neural Information Processing Systems*, vol. 36, 2023.
- [C8] **[IEEE CDC] Zhang, Hongchao**, Zhouchi Li, Hongkai Dai, and Andrew Clark. "Efficient Sum of Squares-Based Verification and Construction of Control Barrier Functions by Sampling on Algebraic Varieties." In *IEEE Conference on Decision and Control*, 5384–5391, 2023.
- [C9] **[VehicleSec] Zhang, Hongchao**, Zhouchi Li, Shiyu Cheng, and Andrew Clark. "Cooperative Perception for Safe Control of Autonomous Vehicles under LiDAR Spoofing Attacks." In *ISOC Symposium on Vehicle Security and Privacy*, 2023. **GM AutoDriving Security Award.**
- [C10] **[IEEE CDC] Zhang, Hongchao**, Shiyu Cheng, Luyao Niu, and Andrew Clark. "Barrier Certificate based Safe Control for LiDAR-based Systems under Sensor Faults and Attacks." In *IEEE Conference on Decision and Control*, 2256–2263, 2022.
- [C11] **[IEEE CDC] Niu, Luyao, Hongchao Zhang**, and Andrew Clark. "Safety-Critical Control Synthesis for Unknown Sampled-Data Systems via Control Barrier Functions." In *IEEE Conference on Decision and Control*, 6806–6813, 2021.
- [C12] **[IEEE ICRA] Zhang, Hongchao**, Zhouchi Li, and Andrew Clark. "Model-based Reinforcement Learning with Provable Safety Guarantees via Control Barrier Functions." In *IEEE International Conference on Robotics and Automation*, 792–798, 2021.
- [C13] **[IEEE CDC] Clark, Andrew, Zhouchi Li, and Hongchao Zhang**. "Control Barrier Functions for Safe CPS Under Sensor Faults and Attacks." In *IEEE Conference on Decision and Control*, 796–803, 2020.

AWARDS & HONORS

- General Motors AutoDriving Security Award, ISOC Symposium on Vehicle Security (VehicleSec @ NDSS), 2023
- Shao-Fang and Tsu-Chin Lee Endowed Fellowship, Washington University in St. Louis, 2022

RESEARCH PROJECTS

Anomaly Detection for Nashville Water Distribution Network	2025–Present
Developing anomaly detection frameworks for Nashville metro water infrastructure to enable diagnostics and real-time monitoring. Integrating machine learning with physics-based models to detect leakage, valve faults, and pressure anomalies.	
Resilient Safe Control of Networked Systems	2025–Present
Developing online safe control frameworks for network systems under potential faults and cyber attacks. Integrating physics models and formal method to ensure safe and stable operation that scales up for network systems.	
Neural Symbolic Safe Control of AI-enabled Systems IEEE TAC 2025, NeurIPS 2023, 2024, ICRA 2021	2019–Present
Developed the first exact neural safety certificate and online safe control via ReLU neural control barrier functions Integrated AI/ML with control theory to resolve theoretical gaps in safety verification and achieve 100x scalability improvement. Provided neural-symbolic verification that supported my advisor's NSF DCSD grant CMMI-2418806.	
Synthesis and Control of Cyber-Resilient CPS IEEE TAC 2025, ICRA 2024, CDC 2020	2019–2025
Pioneered fault-tolerant control providing joint safety and stability guarantees under sensor faults, and adversarial attacks. Utilize control theory to develop fault detection and recovery frameworks for LiDAR sensors with safety guarantees. Received GM AutoDriving Security Award for contributions to autonomous vehicle security.	

TEACHING

Guest Lecturer: VU CS 3892/5892: Projects in Cloud Computing (2025 Fall), VU CS 6315: Automated Verification (2026 Spring)

Assistant Instructor: WashU ESE 559: Learning and Planning in Robotics (2025 Spring)

Leading Assistant Instructor: WashU ESE 2180 Linear Algebra and Component Analysis (2024 Fall)

MENTORING

Vanderbilt University Graduate Directed Research: Yilin Liu (2025)

Vanderbilt University Undergraduates Directed Research: Aarij Atiq (2025)

Washington University Research Experience for Undergraduates (REU):

Ziyi Peng (2025); Mario Rodriguez-Montoya, Dzenan Zecevic (2024); Nicolas Hernandez, Keyron Linarez (2023)

Washington University Graduate Directed Research: Mark Charnot, Seifeldin I. Elkhatab (2023)

Worcester Polytechnic Institute Major Qualifying Project (MQP):

Noelle Johnson (2022); Minh Le and Christopher Letherbarrow (2019)

SERVICE

AAAI (Association for the Advancement of Artificial Intelligence) Reviewer 2026

NeurIPS (Advances in Neural Information Processing Systems) Reviewer 2025

IEEE TCNS (IEEE Transactions on Control of Network Systems) Reviewer 2025, 2026

IEEE TAC (IEEE Transactions on Automatic Control) Reviewer 2023, 2024, 2025, 2026

IEEE CDC (IEEE Conference on Decision and Control) Reviewer 2022, 2023, 2024, 2025

IEEE T-IFS (IEEE Transactions on Information Forensics and Security) Reviewer 2023, 2024, 2025

ACM TCPS (ACM Transactions on Cyber-Physical Systems) Reviewer 2023, 2024, 2025

IEEE TRO (IEEE Transactions on Robotics) Reviewer 2025

IEEE L-CSS (IEEE Robotics and Automation Letters) Reviewer 2025